Harikeshav Rameshkumar

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Research Interests

Efficient and scalable machine learning systems; quantized inference; heterogeneous and distributed computing; compiler/runtime design for private inference.

Education

The Ohio State University Columbus, OH
B.S. in Computer Science; GPA: 3.9/4.0 *Expected May 2028*

University Honors; Dean's List (all semesters).
Selected coursework: Operating Systems, Systems Programming, Data Structures & Algorithms, Computer Networking, Principles of Programming Languages, Database Systems, Introduction to Deep Learning & AI.

Research Experience

Indian Institute of Technology Madras Remote
Research Intern *Jun. 2023 – Oct. 2023*

Machine Learning-Based RSSI Prediction and Heatmap Generation

- Investigated regression models for indoor received signal strength (RSSI) prediction to support wireless coverage mapping.
- Generated 1,000 synthetic spatial samples using a log-distance path-loss model; compared decision trees, Gaussian process regression (GPR), KNN, linear regression, and SVR using a common 80/20 train–test split.
- Evaluated R^2, MAE, MSE, and RMSE; GPR achieved $R^2 = 0.975$, MAE $= 0.282$ dB, and RMSE $= 0.931$ dB on the synthetic test split, followed by KNN ($R^2 = 0.961$).

Selected Research & Systems Projects

Penumbra-FHE — Encrypted ML Inference Python, Rust, ONNX

- Investigated latency–memory–fidelity tradeoffs between TFHE and leveled CKKS, controlling graph structure and quantization to distinguish backend behavior from toolchain differences.
- Developed a Python/Rust framework with a shared quantized IR and integer reference; evaluated 14 workloads across 13 distinct graphs, measuring latency, peak memory, and prediction agreement.
- Demonstrated that error-bound compliance need not preserve predictions: CKKS met its declared bound yet flipped 80.28% of Branched Digits labels; TFHE matched the reference on 30 checked samples per workload.
- Traced over 90% of TFHE bootstraps in evaluated CNNs to residual carry/primitive-lowering operations; characterized CKKS's lower latency on 11 of 12 supported workloads alongside higher memory demand.

LEAP — Heterogeneous Distributed LLM Inference C++20, C, SIMD

- Explored transformer inference across heterogeneous, memory-constrained machines through a C++20 runtime with asymmetric layer partitioning and activation exchange over a logical Ethernet ring.
- Examined quantization and data movement as design constraints; implemented blockwise INT8 weights, memory-mapped loading, and AVX2/FMA and ARM NEON kernels for cross-architecture inference.
- Developed TCP/UDP and Netfilter/shared-memory backends to explore communication-path design, with separate prefill/decode protocols and configurable layer boundaries for heterogeneous execution.

Manuscripts & Research Reports

1. **Harikeshav Rameshkumar**. *A Controlled Comparison of tfhe-rs Radix-Integer and poulpy-ckks Inference under a Shared Quantized IR.* Unpublished manuscript draft.
2. **Harikeshav Rameshkumar**, Venkatesh Ramaiyan. *Machine Learning-Based RSSI Prediction and Heatmap Generation.* Research report draft, IIT Madras internship.

Selected Industry Experience

GE Aerospace Bengaluru, India
Software Engineering Intern, AI/FinFlow Team *Jun. 2026 – Aug. 2026*

- Developed LLM-based contract-extraction pipelines using AWS Bedrock and Claude; reduced per-document processing time from 1,957 to 505 seconds through pipeline optimization, streaming, rate limiting, and cached page splitting.
- Built a FastAPI/Amazon Neptune assistant for querying financial-lineage data; expanded tests from 29 to 78 and eliminated 93 strict type-checking errors across 31 modules.
- Built a Python/Polars/DuckDB analytics pipeline processing 726K+ logs in 15 seconds; traced and corrected a defect that inflated reported adoption figures.

Siage Solutions Bengaluru, India
Software Engineering Intern *Jun. 2025 – Aug. 2025*

- Built a retrieval-augmented question-answering system over 100+ technical manuals (15M+ tokens) using LangChain, Qdrant, and OpenAI embeddings; reduced reported search time from 4.2 seconds to 350 milliseconds.
- Developed ticket triage using fine-tuned BERT and a vector index; reduced duplicate tickets by 42% and resolution time from 14.2 to 4.0 hours.
- Deployed a vLLM-based sentiment service on AWS EKS, increasing reported token throughput by 4.8×.

Additional Experience

Wayfair — AI Automation Apprenticeship Dec. 2025 – Feb. 2026
Built automated trend-research and competitive-intelligence workflows using n8n.

Palantir Technologies — Technical Fellow Nov. 2025 – Jan. 2026
Built Foundry ingestion, transformation, Ontology, and interactive data-exploration workflows.

Selected Honors & Awards

- **2nd Place, Medpace Track**, RevolutionUC 2026 — Pulse, multi-agent clinical-trial safety platform.
- **3rd Place, Visa Track**, TartanHacks 2026 — Penny, AI personal-finance assistant.
- **3rd Place, The Token Company Track**, NextHacks 2026 — Distill, LLM context-compression engine.
- **Best AI Hack Runner-Up**, HackOHI/O 2025 — LeadForge, agentic sales platform.

Technical Skills

Languages: Python, C++, Rust, C, SQL.
ML & numerical computing: PyTorch, TensorFlow, scikit-learn, ONNX, quantization, model calibration, regression, transformer inference.
Systems & compilation: SIMD (AVX2/FMA, ARM NEON), OpenMP, Linux kernel modules, TCP/UDP, memory mapping, intermediate representations, graph evaluation.
Private inference & evaluation: tfhe-rs, CKKS, cleartext-reference testing, latency/memory benchmarking, pytest, CI/CD.
Data & infrastructure: Polars, DuckDB, Qdrant, AWS, Docker, Kubernetes.